

Definite integrals – 3

Antiderivatives and the Newton–Leibniz formula

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Theoretical minimum.

Theorem 1 (the integral as a function of its upper limit). If f is integrable on $[a, b]$, then $F(x) = \int_a^x f$ is Lipschitz with constant $\sup |f|$; F is differentiable at every point of continuity of f and $F' = f$ there. At a jump of f the one-sided derivatives of F exist and equal the one-sided limits of f .

Theorem 2 (Newton–Leibniz). If f is integrable on $[a, b]$ and Φ is any antiderivative of f on $[a, b]$, then $\int_a^b f = \Phi(b) - \Phi(a)$. Both hypotheses are needed and neither implies the other: there are integrable functions with no antiderivative and functions with an antiderivative which are not integrable.

Fact 1 (why the two classes differ). A derivative has the Darboux property, so a function with a jump has no antiderivative although it may be integrable. Conversely Volterra's construction gives a differentiable F whose derivative is bounded but not Riemann integrable, so the formula of Theorem 2 cannot be applied to F' .

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1. Prove Theorem 1 and Theorem 2. Give an f for which F is differentiable at a point where f is discontinuous, and an f for which F fails to be differentiable on a dense set.
 2. Prove that every derivative has the Darboux property and deduce that a function with a jump discontinuity has no antiderivative. Conclude that the class of integrable functions and the class of functions with antiderivatives are incomparable.
 3. (MIPT) Prove that the function equal to $\sin \frac{1}{x}$ for $x \neq 0$ and to 0 at $x = 0$ has an antiderivative on $\mathbb{R}$.

4. (RMO) Let $a > 0$ and let $f: [0, \infty) \rightarrow [0, a]$ be continuous on $(0, \infty)$ and have the Darboux property on $[0, \infty)$. Prove that if $f(0) = 0$ and $xf(x) \geq \int_0^x f$ for all $x \geq 0$, then f admits antiderivatives on $[0, \infty)$.
5. Prove that the set of functions admitting antiderivatives is closed under addition but not under multiplication, and construct a function which admits an antiderivative on $[0, 1]$ yet is unbounded on every subinterval, hence Riemann integrable on none of them. Explain how this coexists with Theorem 2.